Estimating in Building Construction

Chapter 5 Computers in Estimating

Benefits of Computers

- Reduction of calculation errors
- Increased estimating speed
- Many computerized estimating packages allow the estimator to track where quantities came from
- Quick and easy to make changes

Computerized Estimating

- Computer cannot think for the estimator.
- Computerized estimating is a tool, just like a calculator
- Always ask, Does this number make sense?

Dangers of Computers

- Turn thinking over to computer
  - Estimating becomes data-entry
- Using the software for estimates it was not designed for
  - Results in incorrect quantities, costs, labor hours, etc.

Spreadsheets software (Microsoft Excel)

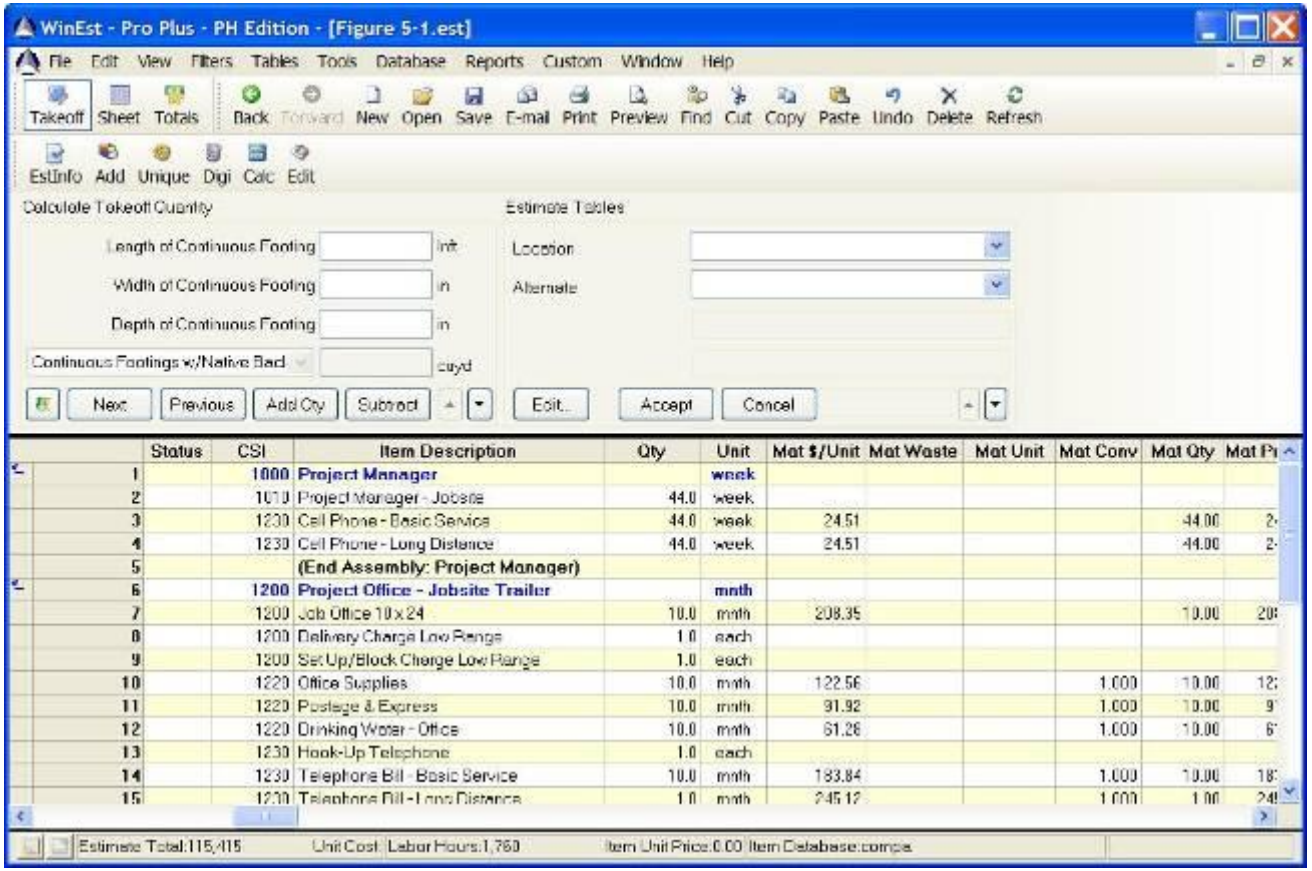
- Often used in conjunction with specialized software packages
- In 2003, the American Society of Professional Estimators reported that 29% of construction companies used Excel as an estimating tool.
- Advantages
  - Cheap
  - Easily to adapted

Specialized Estimating Software

- Advantages:
  - Can take off assemblies
  - Easily viewed in different formats
  - Prepares standard and custom reports
  - Can quickly make changes

Specialized Estimating Software

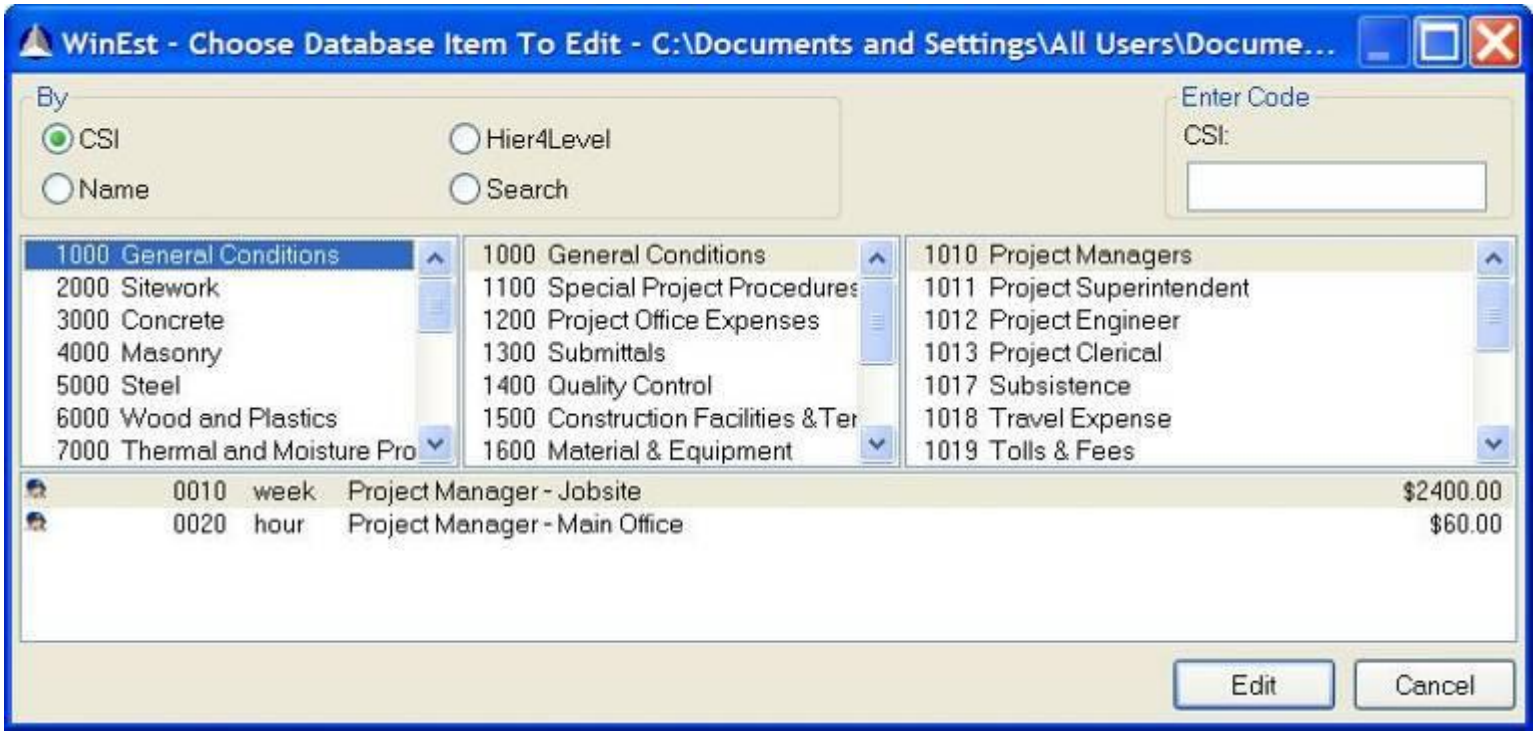
- Disadvantages
  - Expensive
  - Time consuming to set up and maintain



WinEst Estimate Sheet

Figure 5.1

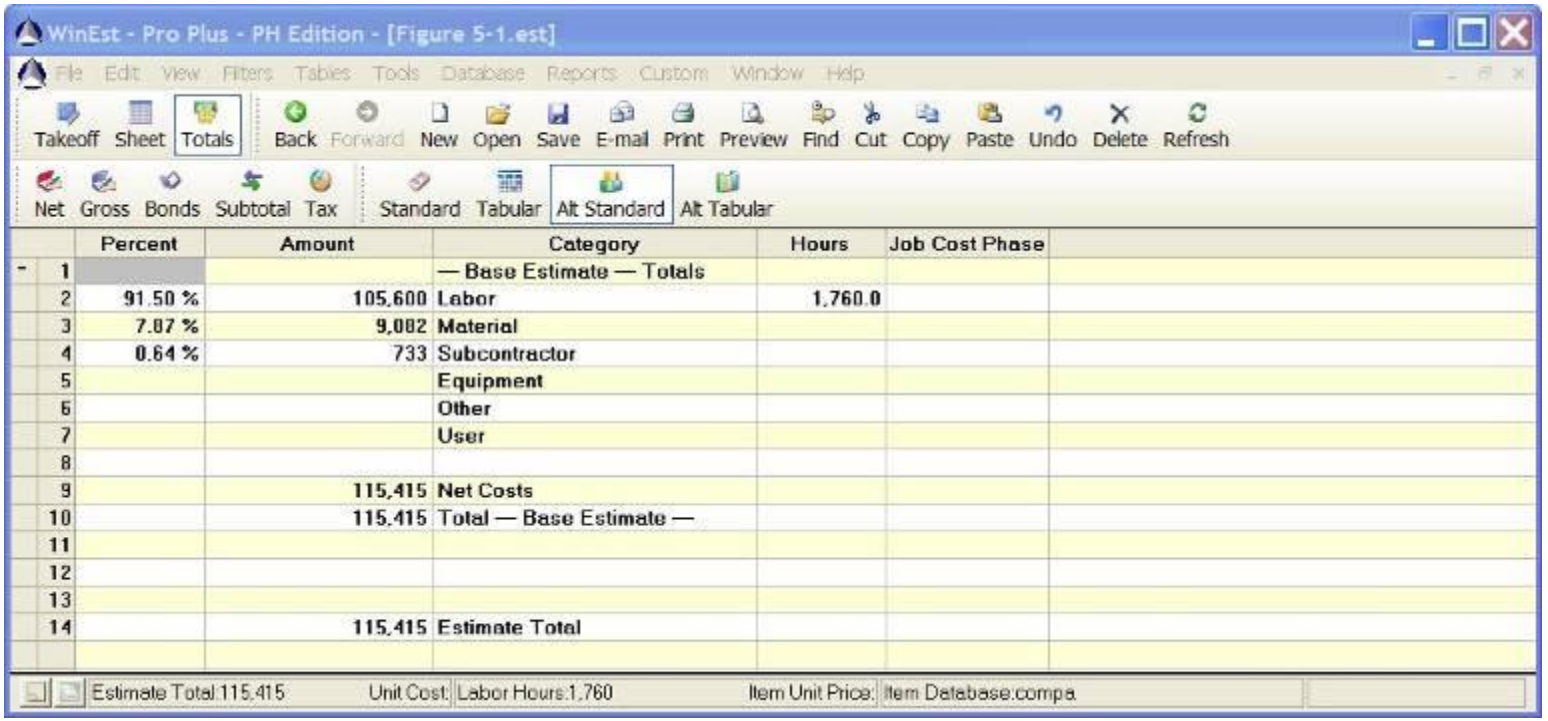
Courtesy of WinEstimator Inc.



WinEst Database

Figure 5.2

Courtesy of WinEstimator Inc.



	Percent	Amount	Category	Hours	Job Cost Phase
1			— Base Estimate — Totals		
2	91.50 %	105,600	Labor	1,760.0	
3	7.07 %	9,082	Material		
4	0.64 %	733	Subcontractor		
5			Equipment		
6			Other		
7			User		
8					
9		115,415	Net Costs		
10		115,415	Total — Base Estimate —		
11					
12					
13					
14		115,415	Estimate Total		

WinEst Totals Page

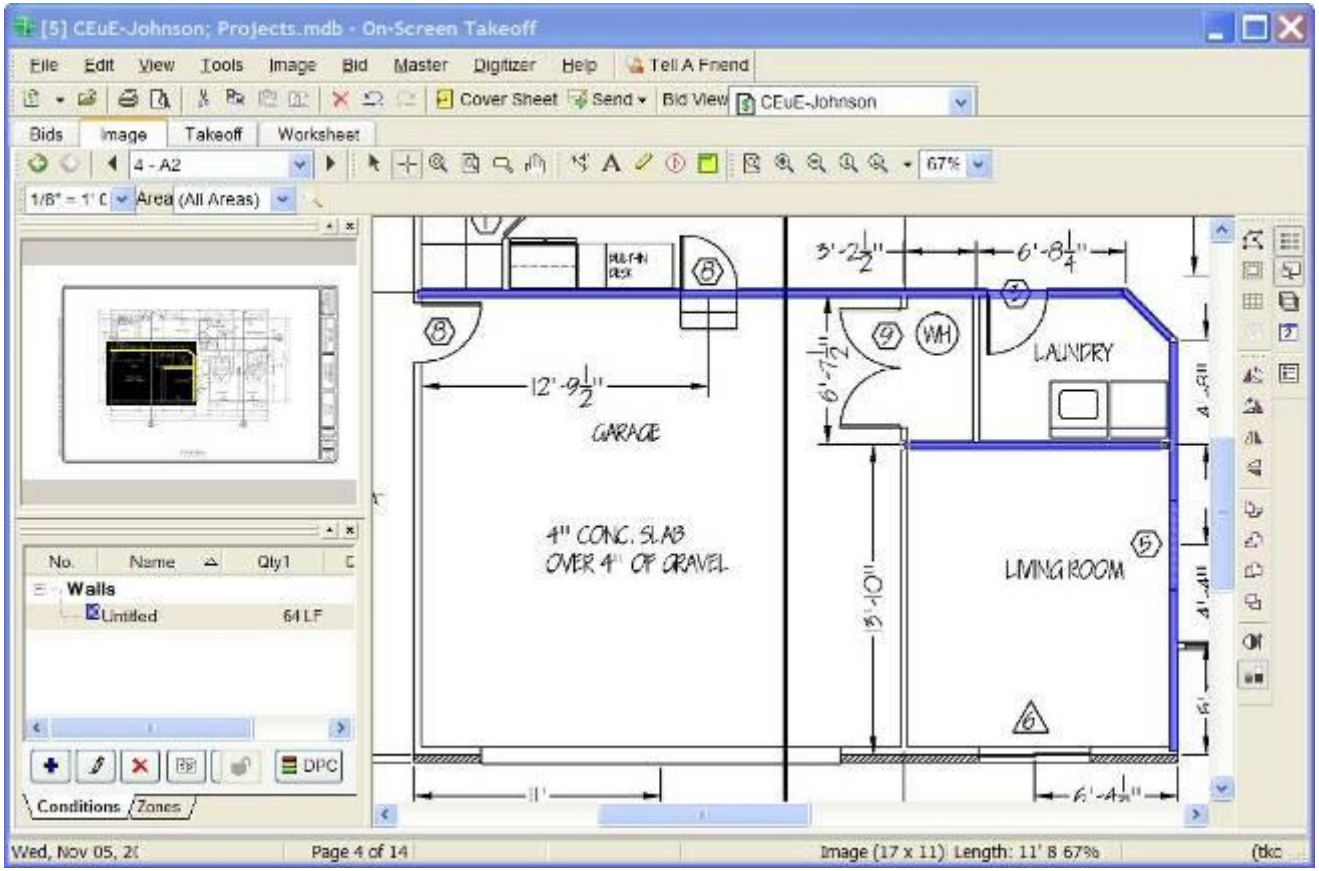
Figure 5.3

Courtesy of WinEstimator Inc.

Takeoff Software

- Determine quantities from electronic set of drawings

- Advantages
  - Track where quantities come from
  - Calculate areas and volumes
  - Inexpensive to reproduce drawings
    - Can be emailed



On-Screen Takeoff

Figure 5.4

Courtesy of OnCenter Software

Read Chapter 1, 2, 3, 4

- Book
  - Peurifoy, R.L. and Oberlender, G.D. (2002). Estimating Construction Costs. McGraw-Hill.

Thank You